

Algebra: Factor & Remainder Theorem

Practice Worksheet • numberbender.com

N

Name: _____ Date: _____

Score: / 10

DIRECTIONS

Use the Remainder Theorem to evaluate $P(c)$. Use the Factor Theorem to test factors.

- 1 Use Remainder Theorem. Find $P(3)$:

$$P(x) = 3x^2 + 3x + 4$$

Answer: _____

- 2 Use Remainder Theorem. Find $P(3)$:

$$P(x) = 3x^2 + 1x - 5$$

Answer: _____

- 3 Use Remainder Theorem. Find $P(3)$:

$$P(x) = 3x^2 + 2x + 0$$

Answer: _____

- 4 Use Remainder Theorem. Find $P(2)$:

$$P(x) = 2x^2 - 2x - 1$$

Answer: _____

- 5 Use Remainder Theorem. Find $P(-2)$:

$$P(x) = 2x^2 - 3x + 0$$

Answer: _____

- 6 Is $(x-3)$ a factor?

$$P(x) = x^2 + 2x + 1$$

Answer: _____

- 7 Is $(x-3)$ a factor?

$$P(x) = x^2 - 4x + 3$$

Answer: _____

- 8 Is $(x-3)$ a factor?

$$P(x) = x^2 + 4x + 3$$

Answer: _____

- 9 Is $(x-1)$ a factor?

$$P(x) = x^2 - 4x + 4$$

Answer: _____

- 10 Is $(x-2)$ a factor?

$$P(x) = x^2 - 4$$

Answer: _____



Math is hard. So we made it human.
Labanan ang agwat.

Free to print & share (CC BY 4.0)

Factor and Remainder Theorem Worksheet

More free worksheets, video lessons & courses:

numberbender.com/math-worksheets

Page 1 of 2 · Video: youtu.be/fwkn1JCvIbc



Watch the lesson



More free sheets

Answer Key & Solutions

Algebra: Factor & Remainder Theorem • Numberbender

N

TEACHER NOTES $P(c)$ equals the remainder when $P(x)$ is divided by $(x-c)$. If $P(c)=0$, then $(x-c)$ is a factor.

1 Use Remainder Theorem. Find $P(3)$:

$$= P(3) = 40$$

$$P(x) = 3x^2 + 3x + 4$$

2 Use Remainder Theorem. Find $P(3)$:

$$= P(3) = 25$$

$$P(x) = 3x^2 + 1x - 5$$

3 Use Remainder Theorem. Find $P(3)$:

$$= P(3) = 33$$

$$P(x) = 3x^2 + 2x + 0$$

4 Use Remainder Theorem. Find $P(2)$:

$$= P(2) = 3$$

$$P(x) = 2x^2 - 2x - 1$$

5 Use Remainder Theorem. Find $P(-2)$:

$$= P(-2) = 14$$

$$P(x) = 2x^2 - 3x + 0$$

6 Is $(x-3)$ a factor?

$$= P(3) = 16, \text{ Factor: No}$$

$$P(x) = x^2 + 2x + 1$$

7 Is $(x-3)$ a factor?

$$= P(3) = 0, \text{ Factor: Yes}$$

$$P(x) = x^2 - 4x + 3$$

8 Is $(x-3)$ a factor?

$$= P(3) = 24, \text{ Factor: No}$$

$$P(x) = x^2 + 4x + 3$$

9 Is $(x+1)$ a factor?

$$= P(-1) = 9, \text{ Factor: No}$$

$$P(x) = x^2 - 4x + 4$$

10 Is $(x-2)$ a factor?

$$= P(2) = 0, \text{ Factor: Yes}$$

$$P(x) = x^2 - 4$$

